CHAPTER

6

Angular momentum⍟

OUTLINE

6.1 Commutation relations 220

6.2 Angular momentum algebra 222

6.3 Consequences of angular momentum algebra 230

6.4 Matrix representation of the angular momentum operator 232

6.5 Representation of the angular momentum operator in position space 238

Appendix 6.A Levi-Civita symbol 244

Appendix 6.B Uncertainty principle applied to the angular momentum 244

6.B.1 Schwartz inequality *245*

6.B.2 Schrödinger uncertainty principle *245*

6.B.3 Application: angular momentum *247*

Appendix 6.C Exercises 248

In this chapter, we study the angular momentum, a fundamental physical quantity that plays a crucial role in various phenomena and contributes to the understanding of concepts such as magnetism.

Objective The goal of this chapter is therefore to present the formal algebra of angular momentum (which can be orbital or spin) and its consequences. Understanding this chapter is fundamental to begin the studies in the next chapter, where we will study the addition of angular momenta.

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Quantum Mechanics
https://doi.org/10.1016/B978-0-44-332826-8.00015-4

6.1 Commutation relations

We begin this chapter's journey with the definition of angular momentum:

$$\vec{L} = \vec{r} \times \vec{p}, \tag{6.1}$$

which, obviously, has components:

$$\vec{L} = L_x \hat{x} + L_y \hat{y} + L_z \hat{z}. \tag{6.2}$$

These components can be written in terms of the Levi-Civita symbol ϵ_{ijk} (see Appendix 6.A for more details). Considering further that these quantities are operators, we have:

$$\hat{L}_i = \epsilon_{ijk} \hat{x}_j \hat{p}_k, \tag{6.3}$$

where the subscript index, in this case i, j, k, represents, respectively, the Cartesian components x, y, z. Note also that $\hat{x}_j$ is a generalization of the spatial coordinate, and can be $\hat{x}$, $\hat{y}$, or $\hat{z}$, depending on the value assigned to the associated index. We leave as an exercise the verification that the components of a vector product can be represented as in the above equation.

Our goal in this section is to determine the commutation relations for the angular momentum – and the detailed proof to obtain this result is given in Box 6.1. We therefore have:

$$\boxed{[\hat{L}_u, \hat{L}_v] = i\hbar \epsilon_{uvw} \hat{L}_w} \tag{6.4}$$

which leads to:

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z, \tag{6.5}$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y, \tag{6.6}$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x. \tag{6.7}$$

The result above indicates that the components of angular momentum do not commute with one another. Consequently, it is physically impossible to have simultaneous and precise knowledge of two of these quantities. Mathematically, this means that the matrices representing these components cannot be diagonalized simultaneously, as they cannot share the same vector basis.

Box 6.1 Detailed description of the commutation relations for the angular momenta

To perform the commutator $[\hat{L}_u, \hat{L}_v]$ we begin with Eq. (6.3). We then have:

$$[\hat{L}_u, \hat{L}_v] = [\epsilon_{uab}\hat{x}_a\hat{p}_b, \epsilon_{vcd}\hat{x}_c\hat{p}_d] \\ = \epsilon_{uab}\epsilon_{vcd}[\hat{x}_a\hat{p}_b, \hat{x}_c\hat{p}_d]. \tag{6.8}$$

In order to proceed, we need to use the non-commutation of space and linear momentum.[a] Thus:

$$[\hat{L}_u, \hat{L}_v] = \epsilon_{uab}\epsilon_{vcd}\{\hat{x}_a[\hat{p}_b, \hat{x}_c]\hat{p}_d + \hat{x}_c[\hat{x}_a, \hat{p}_d]\hat{p}_b\} \\ = (i\hbar)\epsilon_{uab}\epsilon_{vcd}\{-\hat{x}_a\hat{p}_d\delta_{bc} + \hat{x}_c\hat{p}_b\delta_{ad}\}. \tag{6.9}$$

Now, each Kronecker delta obtained above acts on the second Levi-Civita symbol:

$$[\hat{L}_u, \hat{L}_v] = (i\hbar)\{-\epsilon_{uab}\epsilon_{vbd}\hat{x}_a\hat{p}_d + \epsilon_{uab}\epsilon_{vca}\hat{x}_c\hat{p}_b\}. \tag{6.10}$$

The indices of the Levi-Civita symbols above can be permuted, respecting the definitions of Eq. (6.A.1):

$$[\hat{L}_u, \hat{L}_v] = (i\hbar)\{\epsilon_{bua}\epsilon_{bvd}\hat{x}_a\hat{p}_d - \epsilon_{aub}\epsilon_{avc}\hat{x}_c\hat{p}_b\}. \tag{6.11}$$

Eq. (6.A.3) can then be applied to the result above, yielding:

$$[\hat{L}_u, \hat{L}_v] = (i\hbar)\{(\delta_{uv}\delta_{ad} - \delta_{ud}\delta_{av})\hat{x}_a\hat{p}_d - (\delta_{uv}\delta_{bc} - \delta_{uc}\delta_{bv})\hat{x}_c\hat{p}_b\} \\ = (i\hbar)(\hat{x}_u\hat{p}_v - \hat{x}_v\hat{p}_u). \tag{6.12}$$

On the other hand, a second Levi-Civita symbol can be added to Eq. (6.3):

$$\epsilon_{uvw}\hat{L}_w = \epsilon_{uvw}\epsilon_{wab}\hat{x}_a\hat{p}_b \\ = (\delta_{ua}\delta_{vb} - \delta_{ub}\delta_{va})\hat{x}_a\hat{p}_b \\ = \hat{x}_u\hat{p}_v - \hat{x}_v\hat{p}_u. \tag{6.13}$$

Therefore, from Eqs. (6.12) and (6.13), we obtain what we are looking for, that is, the commutation rules for the components of angular momentum:

$$[\hat{L}_u, \hat{L}_v] = i\hbar\epsilon_{uvw}\hat{L}_w. \tag{6.14}$$

[a] From Eq. (2.70) we know that $[\hat{x}, \hat{p}] = i\hbar$, this being a relation obtained for a one-dimensional problem. We can generalize this relation to: $[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}$, where the indexes i, j represent the Cartesian components x, y, or z.

6.2 Angular momentum algebra

Since the components of angular momentum do not commute with each other, they cannot be determined simultaneously and with absolute precision.[1] For this reason, we can determine only one of these components. To better understand the angular momentum, we therefore need another operator, but one that commutes with these components. Thus, we consider:

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2. \tag{6.15}$$

Based on the definition above and the results from Section 6.1, it is possible to determine the algebra of the angular momentum.

Theorem 6.1 The operator $\hat{L}^2$ commutes with the components $\hat{L}_u$ of angular momentum:

$$\boxed{[\hat{L}^2, \hat{L}_u] = 0} \tag{6.16}$$

Proof. Considering $u = z$:

$$\begin{aligned}
&[\hat{L}_x\hat{L}_x + \hat{L}_y\hat{L}_y + \hat{L}_z\hat{L}_z, \hat{L}_z] \\
&\quad = [\hat{L}_x\hat{L}_x, \hat{L}_z] + [\hat{L}_y\hat{L}_y, \hat{L}_z] + [\hat{L}_z\hat{L}_z, \hat{L}_z] \\
&\quad = \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x + \hat{L}_y[\hat{L}_y, \hat{L}_z] + [\hat{L}_y, \hat{L}_z]\hat{L}_y \\
&\quad = \hat{L}_x(-i\hbar\hat{L}_y) + (-i\hbar\hat{L}_y)\hat{L}_x + \hat{L}_y(i\hbar\hat{L}_x) + (i\hbar\hat{L}_x)\hat{L}_y \\
&\quad = 0.
\end{aligned} \tag{6.17}$$

This same idea applies to other directions, and it is left as an exercise. ■

Therefore, $\hat{L}^2$ commutes with all components of the angular momentum operator and thus we can simultaneously know the projection of the angular momentum along a given axis, as well as its magnitude. We are free to choose one of the components of angular momentum as a reference. By convention, we choose $\hat{L}_z$. What we will do from now on through theorems, lemmas, and definitions are to obtain eigenvalues and vectors corresponding to these two operators: $\hat{L}^2$ and $\hat{L}_z$.

[1] This concept was discussed for the position-linear momentum $[\hat{x}, \hat{p}] = i\hbar$, with consequences for problems studied throughout this book, such as plane wave and harmonic oscillator.

Definition 6.1 $|a,b\rangle$ are vectors of operators $\hat{L}^2$ and $\hat{L}_z$, being a and b, respectively, their eigenvalues:

$$\hat{L}^2|a,b\rangle = a|a,b\rangle \quad \text{and} \quad \hat{L}_z|a,b\rangle = b|a,b\rangle. \tag{6.18}$$

This definition is appropriate since $\hat{L}_z$ and $\hat{L}^2$ commute. The goal of this subsection is to determine the values of a and b.

Definition 6.2 $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$ are operators, known as raising and lowering.

Based on this definition, we can determine the other components of angular momentum:

$$\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) \quad \text{and} \quad \hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-). \tag{6.19}$$

Because these are observables, the angular momentum operators are Hermitian, while the raising and lowering operators are non-Hermitian.

Theorem 6.2 The raising and lowering operators do not commute with each other:

$$\boxed{[\hat{L}_\pm, \hat{L}_\mp] = \pm 2\hbar\hat{L}_z} \tag{6.20}$$

Proof.

$$\begin{aligned}
[\hat{L}_+, \hat{L}_-] &= [\hat{L}_x + i\hat{L}_y, \hat{L}_x - i\hat{L}_y] \qquad (6.21)\\
&= [\hat{L}_x, \hat{L}_x] - i[\hat{L}_x, \hat{L}_y] + i\left\{[\hat{L}_y, \hat{L}_x] - i[\hat{L}_y, \hat{L}_y]\right\}\\
&= -i[\hat{L}_x, \hat{L}_y] + i[\hat{L}_y, \hat{L}_x]\\
&= -2i[\hat{L}_x, \hat{L}_y]\\
&= 2\hbar\hat{L}_z.
\end{aligned}$$

The case $[\hat{L}_-, \hat{L}_+] = -2\hbar\hat{L}_z$ is left as an exercise for the reader. ■

Theorem 6.3 The component $\hat{L}_z$ of angular momentum do not commute with the raising and lowering operators:

$$\boxed{[\hat{L}_z, \hat{L}_\pm] = \pm\hbar\hat{L}_\pm} \tag{6.22}$$

Proof.

$$
\begin{aligned}
[\hat{L}_z, \hat{L}_+] &= [\hat{L}_z, \hat{L}_x + i\hat{L}_y] \\
&= [\hat{L}_z, \hat{L}_x] + i[\hat{L}_z, \hat{L}_y] \\
&= i\hbar\hat{L}_y + i(-i\hbar)\hat{L}_x \\
&= \hbar\hat{L}_+.
\end{aligned} \tag{6.23}
$$

The case $[\hat{L}_z, \hat{L}_-] = -\hbar\hat{L}_-$ is left as an exercise for the reader. ■

Corollary 6.1 The operator $\hat{L}^2$ commutes with the raising and lowering operators:

$$\boxed{[\hat{L}^2, \hat{L}_\pm] = 0} \tag{6.24}$$

Proof. We know that the raising and lowering operators are linear with $\hat{L}_x$ and $\hat{L}_y$. These, in turn, commute with $\hat{L}^2$, according to Theorem 6.1. Thus, it is natural that $[\hat{L}^2, \hat{L}_\pm] = 0$. ■

Corollary 6.2 The product of raising and lowering operators is related to operators $\hat{L}^2$ and $\hat{L}_z$: $\hat{L}_\pm\hat{L}_\mp = \hat{L}^2 - \hat{L}_z^2 \pm \hbar\hat{L}_z$.

Proof.

$$
\begin{aligned}
\hat{L}_+\hat{L}_- &= (\hat{L}_x + i\hat{L}_y)(\hat{L}_x - i\hat{L}_y) \\
&= \hat{L}_x^2 + \hat{L}_y^2 + i[\hat{L}_y, \hat{L}_x] \\
&= \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z.
\end{aligned} \tag{6.25}
$$

The case $\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$ is left as an exercise for the reader. ■

Theorem 6.4 The raising and lowering operators are non-Hermitian: $\hat{L}_\pm^\dagger = \hat{L}_\mp$.

Proof. The components of the angular momentum operator $\hat{L}_u$, for being observables, are Hermitian, that is, they are complex conjugate transpose ($\hat{L}_u = \hat{L}_u^\dagger$). Thus, since the raising and lowering operators are defined as complex, it is natural that they are non-Hermitian

$$\begin{aligned} \hat{L}_\pm^\dagger &= (\hat{L}_x \pm i\hat{L}_y)^\dagger \\ &= \hat{L}_x^\dagger \mp i\hat{L}_y^\dagger \\ &= \hat{L}_x \mp i\hat{L}_y \\ &= \hat{L}_\mp. \end{aligned} \tag{6.26}$$

■

From now on we have enough elements to achieve the goal of this section: the determination of the eigenvalues and vectors of the operators $\hat{L}_z$ and $\hat{L}^2$. Let's see:

Corollary 6.3 The raising and lowering operators $\hat{L}_\pm$, when applied to $|a,b\rangle$ state, change the quantum number b in one unit of $\hbar$, keeping the quantum number a unchanged: $\hat{L}_\pm|a,b\rangle = c_\pm|a,b\pm\hbar\rangle$.

Proof. To begin, using Theorem 6.3, we analyze the effect of the operator $\hat{L}_z$ on the vector $\hat{L}_\pm|a,b\rangle$:

$$\begin{aligned} \hat{L}_z(\hat{L}_\pm|a,b\rangle) &= \left([\hat{L}_z,\hat{L}_\pm] + \hat{L}_\pm\hat{L}_z\right)|a,b\rangle \\ &= \left(\pm\hbar\hat{L}_\pm + b\hat{L}_\pm\right)|a,b\rangle \\ &= (b\pm\hbar)(\hat{L}_\pm|a,b\rangle). \end{aligned} \tag{6.27}$$

Therefore, notice that the vector $(\hat{L}_\pm|a,b\rangle)$ is a basis vector of the operator $\hat{L}_z$, with eigenvalue $b\pm\hbar$.

Similarly, using Corollary 6.1, we can evaluate the effect of operator $\hat{L}^2$ on the same vector:

$$\begin{aligned} \hat{L}^2(\hat{L}_\pm|a,b\rangle) &= \hat{L}_\pm(\hat{L}^2|a,b\rangle) \\ &= a(\hat{L}_\pm|a,b\rangle), \end{aligned} \tag{6.28}$$

and, as before, we claim that the vector $(\hat{L}_\pm|a,b\rangle)$ is a basis vector of the operator L^2, with eigenvalue a. Note, therefore, that the results above occur exclusively due to the operator $\hat{L}_\pm$, and thus we propose:

$$\hat{L}_\pm|a,b\rangle = c_\pm|a,b\pm\hbar\rangle. \tag{6.29}$$

■

Corollary 6.4 There is a maximum and minimum value for eigenvalue of $\hat{L}_z$: b_{max} and b_{min}, and therefore $\hat{L}_+|a,b_{max}\rangle = \hat{L}_-|a,b_{min}\rangle = 0$.

Proof. We start with the expectation value of operator $\hat{L}^2 - \hat{L}_z^2$ (see Eq. (6.18)):

$$\langle a,b|\left(\hat{L}^2 - \hat{L}_z^2\right)|a,b\rangle = a - b^2. \tag{6.30}$$

However:

$$\langle a,b|\left(\hat{L}^2 - \hat{L}_z^2\right)|a,b\rangle = \langle a,b|\left(\hat{L}_x^2 + \hat{L}_y^2\right)|a,b\rangle \geq 0. \tag{6.31}$$

Therefore, we conclude that $a - b^2 \geq 0$, or, analogously, $a \geq b^2$. From Corollary 6.3, we know that applying $\hat{L}_\pm$ on $|a,b\rangle$ changes the quantum number b in one unit of $\hbar$, while a remains unchanged. Considering $a \geq b^2$, the values of b must be confined between b_{min} and b_{max}. Consequently:

$$\hat{L}_+|a,b_{max}\rangle = \hat{L}_-|a,b_{min}\rangle = 0. \tag{6.32}$$

■

Corollary 6.5 The values of b are confined between b_{max} and $b_{min} = -b_{max}$, and therefore $-b_{max} \leq b \leq +b_{max}$.

Proof. To begin, let's recover the result of Corollary 6.4: $\hat{L}_+|a,b_{max}\rangle = 0$; and the result of Corollary 6.2: $\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$. Thus:

$$\begin{aligned}\hat{L}_-\hat{L}_+|a,b_{max}\rangle &= \left(\hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z\right)|a,b_{max}\rangle \\ &= \left(a - b_{max}^2 - \hbar b_{max}\right)|a,b_{max}\rangle \\ &= 0,\end{aligned} \tag{6.33}$$

and, consequently:

$$a = b_{max}(b_{max} + \hbar). \tag{6.34}$$

Considering $\hat{L}_+\hat{L}_-|a,b_{min}\rangle = 0$, we can verify that:

$$a = b_{min}(b_{min} - \hbar). \tag{6.35}$$

The demonstration of the result above is left to the reader as an exercise. Comparing Eqs. (6.34) and (6.35), we obtain:

$$b_{max}(b_{max} + \hbar) = b_{min}(b_{min} - \hbar), \tag{6.36}$$

hence

$$\hbar(b_{min} + b_{max}) = (b_{min}^2 - b_{max}^2), \tag{6.37}$$

which can be rewritten as follows:

$$(b_{min} - b_{max})(b_{min} + b_{max}) = \hbar(b_{min} + b_{max}). \tag{6.38}$$

The equation above has two solutions. The first one:

$$b_{min} - b_{max} = \hbar \quad \Rightarrow \quad b_{min} = b_{max} + \hbar \tag{6.39}$$

is obviously impossible, and therefore only the second solution is possible:

$$b_{min} + b_{max} = 0 \quad \Rightarrow \quad b_{min} = -b_{max}. \tag{6.40}$$

From the result above: $-b_{max} \leq b \leq +b_{max}$. ∎

Corollary 6.6 $b_{max}/\hbar$ is an integer or half-integer.

Proof. By applying n times the operator $\hat{L}_+$ on the vector $|a, b_{min}\rangle$, we obtain the vector $|a, b_{min} + n\hbar\rangle$, up to a maximum of $|a, b_{max}\rangle$. Thus:

$$b_{max} = b_{min} + n\hbar. \tag{6.41}$$

However, by means of Corollary 6.5, we know that $b_{min} = -b_{max}$, and therefore:

$$b_{max} = -b_{max} + n\hbar \quad \Rightarrow \quad \frac{b_{max}}{\hbar} = \frac{n}{2}. \tag{6.42}$$

Since b is incremented only in units of $\hbar$ (see Corollary 6.3), we conclude that $b_{max}/\hbar = n/2$ is always an integer or half-integer, since n is an integer. ∎

Definition 6.3 Since $n/2$ is an integer or half-integer, we define it as l, which can also assume integer and half-integer values, that is: $n/2 = l$.

Corollary 6.7 The eigenvalues of operator $\hat{L}^2$ are $a = l(l+1)\hbar^2$.

Proof. As a consequence of Definition 6.3, we can recover Eqs. (6.34) and (6.42), and write:

$$\begin{aligned} a &= b_{max}(b_{max} + \hbar) \\ &= l(l+1)\hbar^2. \end{aligned} \tag{6.43}$$

∎

Definition 6.4 Since b increases or decreases in units of $\hbar$ (see Corollary 6.3), we define it as $b = m\hbar$.

Corollary 6.8 The eigenvalues of operator $\hat{L}_z$ are $b = m\hbar$, where $-l \leq m \leq +l$, i.e. $m = -l, -l+1, \cdots, +l-1, +l$.

Proof. From Corollary 6.5, we know that:

$$-b_{max} \leq b \leq +b_{max} \tag{6.44}$$

and from Corollary 6.6 and Definition 6.3, we know that:

$$b_{max} = l\hbar. \tag{6.45}$$

Therefore:

$$b = m\hbar, \tag{6.46}$$

where:

$$-l \leq m \leq +l, \tag{6.47}$$

or, in other words: $m = -l, -l+1, \cdots, +l-1, +l$. It is easy to see that m has $2l+1$ possible values. ∎

We conclude this section with the eigenvalues and vectors of angular momentum, which were determined from theorems and corollaries. In summary:

$$\boxed{\hat{L}^2|l,m\rangle = l(l+1)\hbar^2|l,m\rangle} \tag{6.48}$$

and

$$\boxed{\hat{L}_z|l,m\rangle = m\hbar|l,m\rangle} \tag{6.49}$$

where

$$\boxed{-l \leq m \leq +l \quad \Rightarrow \quad m = -l, -l+1, \cdots, +l-1, +l} \tag{6.50}$$

that is, $2l+1$ possible values of m. Note that above the vectors are being described with their respective quantum numbers l and m. In addition, the above results are for the operators $\hat{L}^2$ and $\hat{L}_z$, which commute with each other. Operators $\hat{L}_x$ and $\hat{L}_y$, which do not commute with $\hat{L}_z$ and therefore are not diagonal in the basis $|l,m\rangle$, can be constructed from Eq. (6.19). Thus, let's proceed with this section and determine $c_\pm$ of Corollary 6.3.

Corollary 6.9 $\hat{L}_\pm|l,m\rangle = \sqrt{l(l+1) - m(m\pm 1)}\hbar|l,m\pm 1\rangle$

These operators are called *raising* and *lowering* because they alter the quantum number m by one unit of $\hbar$.

Proof. From Corollary 6.3 we know that:

$$\hat{L}_\pm|a,b\rangle = c_\pm|a,b\pm\hbar\rangle. \tag{6.51}$$

However, from Eqs. (6.48) and (6.49) we have been writing the basis $|a,b\rangle$ as $|l,m\rangle$, and therefore:

$$\hat{L}_\pm|l,m\rangle = c_\pm|l,m\pm 1\rangle \tag{6.52}$$

and

$$\langle l,m|\hat{L}_\pm^\dagger = \langle l,m\pm 1|c_\pm^*. \tag{6.53}$$

From Eqs. (6.52) and (6.53), it is possible to write:

$$\begin{aligned}\langle l,m|\hat{L}_\pm^\dagger\hat{L}_\pm|l,m\rangle &= \langle l,m\pm 1|c_\pm^* c_\pm|l,m\pm 1\rangle \\ &= |c_\pm|^2.\end{aligned} \tag{6.54}$$

However, from Theorem 6.4 and Corollary 6.2:

$$\hat{L}_\pm^\dagger\hat{L}_\pm = \hat{L}_\mp\hat{L}_\pm = \hat{L}^2 - \hat{L}_z^2 \mp \hbar\hat{L}_z. \tag{6.55}$$

Therefore:

$$\begin{aligned}|c_\pm|^2 &= \langle l,m|\hat{L}_\pm^\dagger\hat{L}_\pm|l,m\rangle \\ &= \langle l,m|\left(\hat{L}^2 - \hat{L}_z^2 \mp \hbar\hat{L}_z\right)|l,m\rangle \\ &= l(l+1)\hbar^2 - m^2\hbar^2 \mp m\hbar^2,\end{aligned} \tag{6.56}$$

and, consequently:

$$|c_\pm| = \sqrt{l(l+1) - m(m\pm 1)}\,\hbar. \tag{6.57}$$

Finally, we conclude that:

$$\boxed{\hat{L}_\pm|l,m\rangle = \sqrt{l(l+1) - m(m\pm 1)}\,\hbar|l,m\pm 1\rangle} \tag{6.58}$$

■

6.3 Consequences of angular momentum algebra

From Eq. (6.4), we know that the components of angular momentum do not commute with each other. This property arises from the commutation relations between the position and linear momentum operators, which, in turn, are directly connected to the uncertainty principle. Therefore, the simultaneous knowledge of two components of angular momentum is impossible. We also deduced Eqs. (6.48) and (6.49), which tell us, respectively, that we can simultaneously know the angular momentum magnitude $\sqrt{l(l+1)}\hbar$ and its projection $m\hbar$ along the z axis. Note also that the projection of angular momentum on the z axis is quantized, being an integer multiple or half-integer multiple of $\hbar$. Despite these results, below we will explore what we know about the components x and y of the angular momentum.

In order to obtain the expectation value of $\hat{L}_x$ and $\hat{L}_y$, Eq. (6.19) is required, which relates these operators with the raising and lowering operators . Thus:

$$\begin{aligned}\langle \hat{L}_x \rangle &= \langle l, m|\hat{L}_x|l, m\rangle \\ &= \frac{1}{2}\Big[\langle l, m|\hat{L}_+|l, m\rangle + \langle l, m|\hat{L}_-|l, m\rangle\Big]\end{aligned} \tag{6.59}$$

and

$$\begin{aligned}\langle \hat{L}_y \rangle &= \langle l, m|\hat{L}_y|l, m\rangle \\ &= \frac{1}{2i}\Big[\langle l, m|\hat{L}_+|l, m\rangle - \langle l, m|\hat{L}_-|l, m\rangle\Big].\end{aligned} \tag{6.60}$$

Nevertheless, according to Eq. (6.58), the raising and lowering operators make the right-hand side of the above equations proportional to:

$$\langle l, m|l, m \pm 1\rangle = 0. \tag{6.61}$$

Thus:

$$\langle \hat{L}_x \rangle = \langle \hat{L}_y \rangle = 0. \tag{6.62}$$

Note that this result is true for any value of $-l \leq m \leq +l$.

We can proceed and calculate the expectation value of the quadratic operator of these components:

$$\langle \hat{L}_x^2 \rangle = \langle l, m|\hat{L}_x^2|l, m\rangle. \tag{6.63}$$

From Eq. (6.19), we can write:

$$\langle \hat{L}_x^2 \rangle = \frac{1}{4}\Big[\langle \hat{L}_+^2 \rangle + \langle \hat{L}_-^2 \rangle + \langle \hat{L}_+\hat{L}_- \rangle + \langle \hat{L}_-\hat{L}_+ \rangle\Big], \tag{6.64}$$

and, for a reason similar to that of Eq. (6.61), the first two terms on the right-hand side of the equation above are null, i.e. $\langle\hat{L}_{\pm}^2\rangle = 0$. However, Corollary 6.2 states that:

$$\hat{L}_{\pm}\hat{L}_{\mp} = \hat{L}^2 - \hat{L}_z^2 \pm \hbar\hat{L}_z \tag{6.65}$$

and, therefore:

$$\langle\hat{L}_{\pm}\hat{L}_{\mp}\rangle = l(l+1)\hbar^2 - m^2\hbar^2 \pm m\hbar^2. \tag{6.66}$$

Thus, we obtain the expectation value of the quadratic operator:

$$\langle\hat{L}_x^2\rangle = \frac{\hbar^2}{2}\left[l(l+1) - m^2\right]. \tag{6.67}$$

It is easy to demonstrate that $\langle\hat{L}_y^2\rangle = \langle\hat{L}_x^2\rangle$.

With these expectation values, it is possible to calculate the variances $\mathrm{var}(L_x)$ and $\mathrm{var}(L_y)$ (Eq. (2.146)):

$$\mathrm{var}(L_x) = \langle\hat{L}_x^2\rangle - \langle\hat{L}_x\rangle^2 = \frac{\hbar^2}{2}\left[l(l+1) - m^2\right], \tag{6.68}$$

which leads us to the standard deviation (Eq. (2.149)):

$$\Delta L_x = \frac{\hbar}{\sqrt{2}}\left[l(l+1) - m^2\right]^{1/2}. \tag{6.69}$$

It's easy to demonstrate that $\Delta L_x = \Delta L_y$. What about ΔL_z? From Eq. (6.49), we observe that $\langle\hat{L}_z\rangle^2 = m^2\hbar^2$ and $\langle\hat{L}_z^2\rangle = m^2\hbar^2$, and therefore $\Delta L_z = 0$.

From the calculations above it is possible to verify that, in fact, we cannot determine with precision the components x and y of the angular momentum, once the component z has been measured. Observe that the standard deviation ΔL_z is zero, which means that we have measured this component and know its value with precision. Due to the uncertainty in determining the components x and y of the angular momentum, we cannot think of this quantity as a vector pointing in a well-defined direction. On the other hand, we can say that the angular momentum resides in the surface area of a cone (with central axis along the z axis and $m\hbar$ height), and any position of the vector on the surface area of this cone is equally probable.

In order to advance this discussion, let's now consider that the projection of the angular momentum on the z axis is maximum, that is $m = l$. In this case, notice that the standard deviation $\Delta L_x = \Delta L_y$ is given by:

$$\Delta L_x(m = l) = \hbar\sqrt{\frac{l}{2}}. \tag{6.70}$$

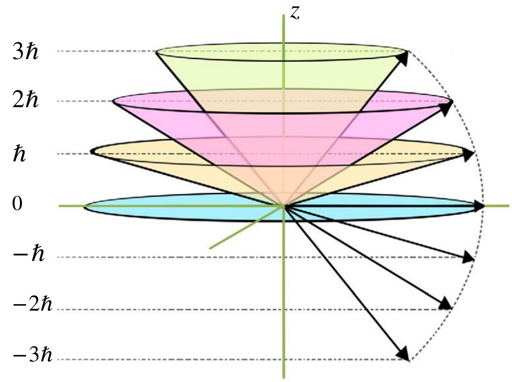

FIGURE 6.1 Schematic representation of an angular momentum $l = 3$. We can measure its module $\sqrt{l(l+1)}\hbar$ and its projection $m\hbar$, but we cannot define its orthogonal components. Therefore, the angular momentum has an equal probability of being found anywhere on the surface of the cone. Furthermore, for the maximum projection value $m = l$, the vector does not lie along the z axis.

Furthermore, for this case, the projection of the vector on the z axis, i.e. $l\hbar$ is smaller than the module $\sqrt{l(l+1)}\hbar$ of the angular momentum. With these arguments we can conclude that, for the case of maximum projection of the angular momentum, it is not absolutely aligned along the z axis. Fig. 6.1, despite being a classical representation of the angular momentum, assists in its visualization.

In Appendix 6.B we will discuss the Schrödinger uncertainty principle (or generalized), applied to the angular momentum. This discussion will complement the ideas presented above.

6.4 Matrix representation of the angular momentum operator

The angular momentum operator is written as discussed in Subsection 3.3. For this purpose, we use Eq. (3.31), where it can be expressed as:

$$\hat{L}_z = \sum_{m'm} \langle l, m'|\hat{L}_z|l, m\rangle |l, m'\rangle\langle l, m|, \tag{6.71}$$

$$\hat{L}^2 = \sum_{m'm} \langle l, m'|\hat{L}^2|l, m\rangle |l, m'\rangle\langle l, m|, \tag{6.72}$$

$$\hat{L}_\pm = \sum_{m'm} \langle l, m'|\hat{L}_\pm|l, m\rangle |l, m'\rangle\langle l, m|, \tag{6.73}$$

where the object $|l, m'\rangle\langle l, m|$ defines the basis vector and

$$\langle l, m'|\hat{L}_z|l, m\rangle = m\hbar\delta_{m',m}, \tag{6.74}$$

$$\langle l, m'|\hat{L}^2|l, m\rangle = l(l+1)\hbar^2\delta_{m',m}, \tag{6.75}$$

$$\langle l, m'|\hat{L}_\pm|l, m\rangle = \sqrt{l(l+1) - m(m\pm 1)}\,\hbar\,\delta_{m',m\pm 1}, \tag{6.76}$$

are the corresponding matrix elements.

We are considering that the system has a single angular momentum value l and, therefore, the eigenvalue m is responsible for identifying the vector. Before discussing the examples of how to construct the matrices for the angular momentum operator, note that this operator contributes with $2l+1$ different states (number of possible values for the eigenvalue m) – this being the dimension of this Hilbert (sub)space. For this reason, the matrices calculated from the results above have dimension $(2l+1)\times(2l+1)$.

■ **Example 6.1**

In this example, we construct the matrix representation of the angular momentum operators $\hat{L}_x$, $\hat{L}_y$ and $\hat{L}_z$, for the case $l = 1$. We begin by defining the basis on which these operators are constructed. Let's consider the vectors $|l, m\rangle$ written as in Example 3.1 and in Eq. (3.32). Thus:

$$|1,+1\rangle = \begin{pmatrix}1\\0\\0\end{pmatrix} \quad |1,0\rangle = \begin{pmatrix}0\\1\\0\end{pmatrix} \quad |1,-1\rangle = \begin{pmatrix}0\\0\\1\end{pmatrix}. \tag{6.77}$$

Verify that, in fact, $\sum_m |l, m\rangle\langle l, m| = \mathbb{1}$. To obtain the operator $\hat{L}_z$, we considered Eqs. (6.71) and (6.74), yielding:

$$\hat{L}_z = \sum_m m\hbar|l, m\rangle\langle l, m|, \tag{6.78}$$

and, consequently:

$$\begin{aligned}\hat{L}_z &= (+\hbar)|1,+1\rangle\langle 1,+1| + 0|1,0\rangle\langle 1,0| + (-\hbar)|1,-1\rangle\langle 1,-1| \\ &= \hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&0\end{pmatrix} + 0\begin{pmatrix}0&0&0\\0&1&0\\0&0&0\end{pmatrix} - \hbar\begin{pmatrix}0&0&0\\0&0&0\\0&0&1\end{pmatrix} \\ &= \hbar\begin{pmatrix}1&0&0\\0&0&0\\0&0&-1\end{pmatrix}.\end{aligned} \tag{6.79}$$

To obtain $\hat{L}_x$ and $\hat{L}_y$, we have to determine the raising and lowering operators, which can be written as:

$$\hat{L}_\pm = \sum_m \sqrt{l(l+1) - m(m\pm 1)}\,\hbar\,|l, m\pm 1\rangle\langle l, m|. \tag{6.80}$$

Therefore:

$$\begin{aligned}
\hat{L}_+ &= \sqrt{2}\hbar|1,+1\rangle\langle 1,0| + \sqrt{2}\hbar|1,0\rangle\langle 1,-1| \\
&= \sqrt{2}\hbar\left[\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}\right] \\
&= \sqrt{2}\hbar\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (6.81) \\
\hat{L}_- &= \sqrt{2}\hbar|1,0\rangle\langle 1,+1| + \sqrt{2}\hbar|1,-1\rangle\langle 1,0| \\
&= \sqrt{2}\hbar\left[\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}\right] \\
&= \sqrt{2}\hbar\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \qquad (6.82)
\end{aligned}$$

From the operators $\hat{L}_\pm$, we can then obtain the operators $\hat{L}_x$ and $\hat{L}_y$, given by:

$$\hat{L}_x = \frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \text{and} \quad \hat{L}_y = \frac{\hbar i}{\sqrt{2}}\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}. \qquad (6.83)$$

The reader is encouraged to construct the operator $\hat{L}^2$. ■

Let us take a step further. We know from Eq. (6.16) that the operator $\hat{L}^2$ commutes with the three components of the angular momentum and, when studying the algebra of the angular momentum, z was chosen as the reference direction. As studied before, due to the commutation relations for the angular momentum (Eq. (6.4)), once $\hat{L}_z$ has been measured and determined, the other two components cannot be determined simultaneously. However, $\hat{L}_z$ was chosen arbitrarily and, obviously, we could have chosen the components x or y as reference. For this reason, we can predict that the eigenvalues of these two other components, if they were measured instead of $\hat{L}_z$, would also be $m\hbar$. To be clearer, let's take a look at the example below.

■ **Example 6.2**

This is an example similar to that developed in Chapter 3, where the Stern–Gerlach experiment and the postulates of quantum mechanics were discussed. As before, we write the operator matrix $\hat{L}_x$

in the bases $\{|l,m\rangle\}$ and $\{|l,m\rangle_x\}$, as well as determine the change of basis operator. The physical interpretation of these results will also be discussed.

We start with the eigenvalue equation:

$$\hat{L}_x|l,m\rangle_x = \lambda|l,m\rangle_x. \tag{6.84}$$

Note that $\hat{L}_x$ obeys the eigenvalue equation, however, with the basis vector $\{|l,m\rangle_x\}$. For this example, we will consider $l=1$ and the respective operator $\hat{L}_x$, obtained in Eq. (6.83). The eigenvalues are found by setting the determinant of the eigenvalue equation equal to zero in order to avoid a trivial solution. Thus:

$$\begin{vmatrix} -\lambda & \hbar/\sqrt{2} & 0 \\ \hbar/\sqrt{2} & -\lambda & \hbar/\sqrt{2} \\ 0 & \hbar/\sqrt{2} & -\lambda \end{vmatrix} = 0 = -\lambda^3 + 2\lambda\frac{\hbar^2}{2}. \tag{6.85}$$

We therefore find three solutions, which are the eigenvalues:

$$\textbf{(I)}\ \lambda = +\hbar, \quad \textbf{(II)}\ \lambda = 0, \quad \textbf{(III)}\ \lambda = -\hbar. \tag{6.86}$$

Note that the above result is expected, that is, since the direction z was chosen arbitrarily and the eigenvalues of $\hat{L}_z$ are $m\hbar$, these are the eigenvalues for any arbitrary direction u. We will now obtain the vectors of $\hat{L}_x$ – obviously, one vector for each eigenvalue.

(I) Eigenvalue $\lambda = +\hbar$. From the eigenvalue Eq. (6.84), it is possible to determine the vector:

$$|1,+1\rangle_x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \tag{6.87}$$

that is:

$$\begin{pmatrix} -\hbar & \hbar/\sqrt{2} & 0 \\ \hbar/\sqrt{2} & -\hbar & \hbar/\sqrt{2} \\ 0 & \hbar/\sqrt{2} & -\hbar \end{pmatrix}\begin{pmatrix} a \\ b \\ c \end{pmatrix} = 0. \tag{6.88}$$

From the first and third rows of this matrix, we determined, respectively: $b=\sqrt{2}a$ and $b=\sqrt{2}c \Rightarrow c=a$. These results, under the vector normalization condition, $|a|^2+|b|^2+|c|^2=1$, lead to $a=1/2$. Thus, the vector for the eigenvalue $\lambda=+\hbar$ reads as:

$$|1,+1\rangle_x = \frac{1}{2}\begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix}. \tag{6.89}$$

(II) Eigenvalue $\lambda = 0$. Analogously to the previous case, the eigenvalue Eq. (6.84) is:

$$\begin{pmatrix} 0 & \hbar/\sqrt{2} & 0 \\ \hbar/\sqrt{2} & 0 & \hbar/\sqrt{2} \\ 0 & \hbar/\sqrt{2} & 0 \end{pmatrix} \begin{pmatrix} a' \\ b' \\ c' \end{pmatrix} = 0. \tag{6.90}$$

From the first row of this matrix, we determine $b' = 0$, while from the second row we determine $c' = -a'$. As in the previous case, these results, under the vector normalization condition lead to $a' = 1/\sqrt{2}$. Thus, the vector for the eigenvalue $\lambda = 0$ reads as:

$$|1, 0\rangle_x = \frac{1}{2} \begin{pmatrix} \sqrt{2} \\ 0 \\ -\sqrt{2} \end{pmatrix}. \tag{6.91}$$

(III) Eigenvalue $\lambda = -\hbar$. As before, we find:

$$\begin{pmatrix} \hbar & \hbar/\sqrt{2} & 0 \\ \hbar/\sqrt{2} & \hbar & \hbar/\sqrt{2} \\ 0 & \hbar/\sqrt{2} & \hbar \end{pmatrix} \begin{pmatrix} a'' \\ b'' \\ c'' \end{pmatrix} = 0. \tag{6.92}$$

From the first row of this matrix we determine $b'' = -\sqrt{2}a''$, while from the third row we determine $b'' = -\sqrt{2}c'' \Rightarrow c'' = a''$. Considering the vector normalization condition we obtain $a'' = 1/2$. Thus, the vector for the eigenvalue $\lambda = -\hbar$ reads as:

$$|1, -1\rangle_x = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{2} \\ 1 \end{pmatrix}. \tag{6.93}$$

The vectors $|l, m\rangle$ can be associated with the vectors $|l, m\rangle_x$. For the case of the vector $|1, +1\rangle_x$ (Eq. (6.89)), it is possible to write:

$$|1, +1\rangle_x = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{2} \\ 1 \end{pmatrix} = \frac{1}{2} \left[\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \sqrt{2} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right]. \tag{6.94}$$

In a more compact way:

$$|+1\rangle_x = \frac{1}{2}|+1\rangle + \frac{\sqrt{2}}{2}|0\rangle + \frac{1}{2}|-1\rangle. \tag{6.95}$$

For simplicity, we omitted the quantum number l, since we are studying only a single angular momentum. The other two vectors can also be represented in terms of the vectors $|m\rangle$, as follows:

$$|0\rangle_x = \frac{\sqrt{2}}{2}|+1\rangle - \frac{\sqrt{2}}{2}|-1\rangle, \tag{6.96}$$

$$|-1\rangle_x = \frac{1}{2}|+1\rangle - \frac{\sqrt{2}}{2}|0\rangle + \frac{1}{2}|-1\rangle. \tag{6.97}$$

Eqs. (6.95), (6.96) and (6.97) can be summarized as follows:

$$\begin{pmatrix} |+1\rangle_x \\ |0\rangle_x \\ |-1\rangle_x \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix} \begin{pmatrix} |+1\rangle \\ |0\rangle \\ |-1\rangle \end{pmatrix} \tag{6.98}$$

or, in a simpler way:

$$|m\rangle_x = \hat{U}|m\rangle, \tag{6.99}$$

where $\hat{U}$ is the change of basis matrix. That is, it takes the system from basis $\{|m\rangle\}$ to basis $\{|m\rangle_x\}$, and vice versa. Observe that vectors $|m\rangle_x$ were written above in terms of vectors $|m\rangle$, but the inverse is also possible:

$$|m\rangle = \hat{U}^{-1}|m\rangle_x. \tag{6.100}$$

Remember that a unitary operator must satisfy the condition $\hat{U}^{-1} = \hat{U}^\dagger$ and consequently: $\hat{U}^\dagger\hat{U} = \mathbb{1}$. We verify that the condition $\hat{U}^\dagger\hat{U} = \mathbb{1}$ is satisfied and thus:

$$\hat{U}^{-1} = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}. \tag{6.101}$$

With the matrix $\hat{U}$ determined, we can diagonalize the operator L_x (non-diagonal in basis $\{|l,m\rangle\}$). Thus, by using Eq. (3.A.6), it is possible to write:

$$(\hat{L}_x)_x = \hat{U}(\hat{L}_x)_z\hat{U}^\dagger \tag{6.102}$$

which, in matrix terms, reads as:

$$(L_x)_x = \left[\frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}\right] \times \left[\frac{\hbar}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\right]\left[\frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}\right], \tag{6.103}$$

and consequently:

$$(\hat{L}_x)_x = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{6.104}$$

The matrix is diagonal, as observed, and has the same eigenvalues determined earlier. This operation corresponds to a rotation of the axes, where the x-axis now takes on the physical interpretation previously associated with the z-axis. ■

6.5 Representation of the angular momentum operator in position space

The goal of this subsection is to describe the angular momentum operators in position space. Let's begin this task by considering the definition of the angular momentum $\vec{\hat{L}} = \vec{r} \times \vec{\hat{p}}$. Our goal is a relatively straightforward task since we know the differential formulation of the operator $\vec{\hat{p}}$, i.e. $\vec{\hat{p}} = -i\hbar\vec{\nabla}$. Therefore:

$$\vec{\hat{L}} = (-i\hbar)\vec{r} \times \vec{\nabla}. \tag{6.105}$$

Considering that (see Appendix 5.B):

$$\vec{\nabla} = \hat{r}\frac{\partial}{\partial r} + \frac{\hat{\theta}}{r}\frac{\partial}{\partial \theta} + \frac{\hat{\phi}}{r\sin\theta}\frac{\partial}{\partial \phi}, \tag{6.106}$$

we can then focus on this vector product:

$$\vec{r} \times \vec{\nabla} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{\phi} \\ r & 0 & 0 \\ \frac{\partial}{\partial r} & \frac{1}{r}\frac{\partial}{\partial \theta} & \frac{1}{r\sin\theta}\frac{\partial}{\partial \phi} \end{vmatrix} = \hat{\phi}\frac{\partial}{\partial \theta} - \frac{\hat{\theta}}{\sin\theta}\frac{\partial}{\partial \phi}. \tag{6.107}$$

With these simple results we conclude that:

$$\boxed{\vec{\hat{L}} = (-i\hbar)\left(\hat{\phi}\frac{\partial}{\partial \theta} - \frac{\hat{\theta}}{\sin\theta}\frac{\partial}{\partial \phi}\right)} \tag{6.108}$$

! It is important to clarify that the symbol ˆ is used to denote both unit vectors and operators, which, in this section, appear together in the same equation. Nevertheless, we have maintained this notation to preserve consistency throughout the book. Readers should find it straightforward to distinguish between unit vectors and operators based on the context within the section.

Obtaining the angular momentum components from the above equations is also a relatively straightforward task, since the component along the u direction can be written as $\hat{L}_u = \hat{u} \cdot \vec{\hat{L}}$, where $\hat{u}$ is a unit vector along the u-direction. Appendix 5.B describes in detail the spherical coordinates, including the unit vectors along the x, y and z directions:

$$\hat{x} = \hat{r}\sin\theta\cos\phi + \hat{\theta}\cos\theta\cos\phi - \hat{\phi}\sin\phi, \tag{6.109}$$

$$\hat{y} = \hat{r}\sin\theta\sin\phi + \hat{\theta}\cos\theta\sin\phi + \hat{\phi}\cos\phi, \tag{6.110}$$

$$\hat{z} = \hat{r}\cos\theta - \hat{\theta}\sin\theta. \tag{6.111}$$

Therefore:

$$\hat{L}_z = \hat{z} \cdot \vec{\hat{L}} = (-i\hbar)\left(\hat{r}\cos\theta - \hat{\theta}\sin\theta\right) \cdot \left(\hat{\phi}\frac{\partial}{\partial\theta} - \frac{\hat{\theta}}{\sin\theta}\frac{\partial}{\partial\phi}\right), \tag{6.112}$$

which can be summarized as:

$$\boxed{\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}} \tag{6.113}$$

In a similar fashion:

$$\hat{L}_x = \hat{x} \cdot \vec{\hat{L}} = (-i\hbar)\left(\hat{r}\sin\theta\cos\phi + \hat{\theta}\cos\theta\cos\phi - \hat{\phi}\sin\phi\right) \cdot \left(\hat{\phi}\frac{\partial}{\partial\theta} - \frac{\hat{\theta}}{\sin\theta}\frac{\partial}{\partial\phi}\right). \tag{6.114}$$

Thus:

$$\boxed{\hat{L}_x = i\hbar\left(\cot\theta\,\cos\phi\frac{\partial}{\partial\phi} + \sin\phi\frac{\partial}{\partial\theta}\right)} \tag{6.115}$$

Finally:

$$\hat{L}_y = \hat{y} \cdot \vec{\hat{L}} = (-i\hbar)\left(\hat{r}\sin\theta\sin\phi + \hat{\theta}\cos\theta\sin\phi + \hat{\phi}\cos\phi\right) \cdot \left(\hat{\phi}\frac{\partial}{\partial\theta} - \frac{\hat{\theta}}{\sin\theta}\frac{\partial}{\partial\phi}\right), \tag{6.116}$$

and therefore:

$$\boxed{\hat{L}_y = i\hbar\left(\cot\theta\,\sin\phi\frac{\partial}{\partial\phi} - \cos\phi\frac{\partial}{\partial\theta}\right)} \tag{6.117}$$

With these three components, we can now obtain the raising $\hat{L}_+ = \hat{L}_x + i\hat{L}_y$ and lowering $\hat{L}_- = \hat{L}_x - i\hat{L}_y$ operators in differential form :

$$\boxed{\hat{L}_\pm = \hbar e^{\pm i\phi}\left(\pm\frac{\partial}{\partial\theta} + i\,\cot\theta\,\frac{\partial}{\partial\phi}\right)} \tag{6.118}$$

Considering we have all the operators above, it is easy to calculate the square of the angular momentum operator $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$, given by:

$$\boxed{\hat{L}^2 = -\hbar^2\left[\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]} \tag{6.119}$$

or, alternatively:

$$-\frac{\hat{L}^2}{\hbar^2} = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}. \tag{6.120}$$

Let's save the result above and recover the equation referring to the angular contribution of the separation of variables when we studied problems with central potentials (Subsection 5.2), more precisely Eq. (5.12):

$$-\lambda\Theta(\theta)\Phi(\phi) = \frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)\Phi(\phi)\Theta(\theta) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\Phi(\phi)\Theta(\theta), \tag{6.121}$$

where it is known, from Eq. (5.33), that:

$$Y_l^m(\theta,\phi) = A_{lm}\Theta(\theta)\Phi(\phi) \tag{6.122}$$

are the spherical harmonics, being A_{lm} the normalization constant given by Eq. (5.34). Furthermore, we know from the development of Subsection 5.2 that $\lambda = l(l+1)$ and $-l \leq m \leq +l$. Therefore, Eq. (6.121) can be rewritten as follows:

$$\left\{\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right\}Y_l^m(\theta,\phi) = -l(l+1)Y_l^m(\theta,\phi). \tag{6.123}$$

Taking Eq. (6.120) into the equation above we obtain:

$$\boxed{\hat{L}^2 Y_l^m(\theta,\phi) = l(l+1)\hbar^2 Y_l^m(\theta,\phi)} \tag{6.124}$$

Therefore, the spherical harmonics are the eigenfunctions of the operator $\hat{L}^2$, and thus these are also the eigenfunctions of the operator $\hat{L}_z$. Thus:

$$\boxed{\hat{L}_z Y_l^m(\theta,\phi) = m\hbar Y_l^m(\theta,\phi)} \tag{6.125}$$

Eqs. (6.124) and (6.125) recover, in position space, the eigenvalue equations for the angular momentum.

■ Example 6.3

Consider a system with angular wave function given by the spherical harmonic:

$$Y_l^m(\theta,\phi) = \frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2(\theta)e^{2i\phi}. \tag{6.126}$$

By utilizing the algebra of angular momentum, we can determine the quantum numbers l and m associated with this state.

The spherical harmonics are eigenfunctions of the operators $\hat{L}^2$ and $\hat{L}_z$ (Eqs. (6.124) and (6.125)). Given the differential formulation of these operators, we can apply them to the eigenfunction under consideration. Thus:

$$\hat{L}_z Y_l^m(\theta,\phi) = -i\hbar\frac{\partial}{\partial\phi}\left(\frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2(\theta)e^{2i\phi}\right) = 2\hbar Y_l^m(\theta,\phi). \tag{6.127}$$

We immediately conclude that $m = +2$. In a similar fashion:

$$\hat{L}^2 Y_l^m(\theta,\phi) = -\hbar^2\left[\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]\left(\frac{1}{4}\sqrt{\frac{15}{2\pi}}\sin^2(\theta)e^{2i\phi}\right) \tag{6.128}$$

which, after a few simple steps, we obtain:

$$\hat{L}^2 Y_l^m(\theta,\phi) = 6\hbar^2 Y_l^m(\theta,\phi). \tag{6.129}$$

Comparing the above equation with Eq. (6.124) we conclude that $l(l+1) = 6$, which implies $l = 2$. Therefore, the eigenfunction of Eq. (6.126) represents the states $l = 2$ and $m = +2$. ■

■ Example 6.4

In this example, we use the algebra of angular momentum to obtain the spherical harmonic corresponding to $(l,m) = (1,-1)$.

We can start solving this problem using Eq. (6.125), which for our case $(l,m) = (1,-1)$ can be written as follows:

$$\hat{L}_z Y_1^{-1}(\theta,\phi) = -\hbar Y_1^{-1}(\theta,\phi). \tag{6.130}$$

We know the differential operator $\hat{L}_z$ (Eq. (6.113)), as well as the fact that spherical harmonics can be written in terms of separation of variables. Thus, Eq. (6.130) can be expressed in the following form:

$$-i\hbar\frac{\partial}{\partial\phi}\Theta(\theta)\Phi(\phi) = -\hbar\Theta(\theta)\Phi(\phi). \tag{6.131}$$

Our goal is to determine the functions $\Theta(\theta)$ and $\Phi(\phi)$. The above equation leads us to a differential equation for $\Phi(\phi)$:

$$\frac{\partial}{\partial\phi}\Phi(\phi) = -i\Phi(\phi), \tag{6.132}$$

which has as a solution:

$$\Phi(\phi) = e^{-i\phi}. \tag{6.133}$$

The next step is to obtain a differential equation for $\Theta(\theta)$. For this task, we can use any angular momentum differential operator that contains the variable θ. However, a bad choice can lead to a differential equation that is difficult to solve. A possible solution can be obtained using the equation:

$$\hat{L}_- Y_1^{-1}(\theta,\phi) = 0, \tag{6.134}$$

which is certainly a differential equation simpler than others (due to the fact that the right-hand side is zero). The lowering operator in differential form was determined and is represented in Eq. (6.118), which leads to:

$$\hbar e^{-i\phi}\left(-\frac{\partial}{\partial\theta} + i\ \cot\theta\ \frac{\partial}{\partial\phi}\right)\Theta(\theta)e^{-i\phi} = 0. \tag{6.135}$$

A few algebraic steps lead to:

$$-\frac{\partial}{\partial\theta}\Theta(\theta) + \cot\theta\,\Theta(\theta) = 0. \tag{6.136}$$

A few more steps (which the reader can easily verify) take the above equation to:

$$\ln[\Theta(\theta)] = \int \cot\theta d\theta = \ln[\sin\theta] \tag{6.137}$$

and, consequently:

$$\Theta(\theta) = \sin\theta. \tag{6.138}$$

Finally, we then have:

$$Y_1^{-1}(\theta,\phi) \propto \sin\theta e^{-i\phi} \quad \Rightarrow \quad Y_1^{-1}(\theta,\phi) = A\sin\theta e^{-i\phi}. \tag{6.139}$$

However, we need to normalize this function (see Eq. (5.46)):

$$\int_0^{\pi}\int_0^{2\pi} |Y_l^m(\theta,\phi)|^2 \sin\theta d\phi d\theta = 1, \tag{6.140}$$

which leads to:

$$|A|^2 \int_0^{\pi} \int_0^{2\pi} \sin^2\theta \sin\theta d\phi d\theta = 1 \quad \Rightarrow \quad A = \sqrt{\frac{3}{8\pi}}, \tag{6.141}$$

and the final result of the problem reads as:

$$Y_1^{-1}(\theta,\phi) = \sqrt{\frac{3}{8\pi}} \sin\theta e^{-i\phi}. \tag{6.142}$$

■

■ **Example 6.5**

Consider that the angular part of a given wave function is given by:

$$A(x,y,z) = \frac{\alpha z}{\sqrt{x^2+y^2+z^2}}, \tag{6.143}$$

where α is a constant. Let's calculate the probability $P(l,m)$ of finding the particle with angular momentum $(l,m) = (1,\pm 1)$ and $(1,0)$.

The first step to solve this problem is to convert the wave function from Eq. (6.143) to spherical coordinates (see Appendix 5.B). We then have:

$$A(\theta,\phi) = \alpha \cos\theta. \tag{6.144}$$

Considering the spherical harmonics, since they are eigenfunctions of the angular momentum operators, it is possible to write (see Eq. (5.39)):

$$A(\theta,\phi) = \alpha \sqrt{\frac{4\pi}{3}} Y_1^0(\theta,\phi). \tag{6.145}$$

We have to determine the constant α and, for this purpose, we use the normalization condition of the (angular) wave function:

$$\int_0^{\pi} \int_0^{2\pi} |A(\theta,\phi)|^2 \sin\theta d\phi d\theta = 1 \quad \Rightarrow \quad \alpha = \sqrt{\frac{3}{4\pi}}. \tag{6.146}$$

Therefore, we obtain the (angular) wave function of the problem:

$$A(\theta,\phi) = Y_1^0(\theta,\phi). \tag{6.147}$$

With this result written in terms of the spherical harmonics we can calculate the probabilities under consideration (see Eq. (2.95)):

$$P(1,\pm 1) = \left| \int_0^{2\pi} \int_0^{\pi} Y_1^{\pm 1*}(\theta,\phi) Y_1^0(\theta,\phi) \sin\theta d\theta d\phi \right|^2 = 0, \tag{6.148}$$

$$P(1,0) = \left| \int_0^{2\pi} \int_0^{\pi} Y_1^{0*}(\theta,\phi) Y_1^0(\theta,\phi) \sin\theta d\theta d\phi \right|^2 = 1. \tag{6.149}$$

To solve the integrals above, we use the orthogonality conditions of the spherical harmonics (Eq. (5.36)). Thus, we conclude from this example that a particle with the proposed wave function has a 100% probability of being in a state characterized by angular momentum $(l,m) = (1,0)$. ■

Appendix 6.A Levi-Civita symbol

The Levi-Civita symbol is a *rank*-3 tensor, since it has 3 indexes. It is defined as:

$$\epsilon_{ijk} = \begin{cases} +1, & \text{for } (i,j,k) \Rightarrow (k,i,j) \Rightarrow (j,k,i) - \text{Even permutations.} \\ -1, & \text{for } (i,j,k) \Rightarrow (i,k,j) - \text{Even permutations.} \\ 0, & \text{for } i=j \text{ or } i=k \text{ or } j=k. \end{cases} \tag{6.A.1}$$

There is a relation between the Levi-Civita symbol and the Kronecker delta:

$$\epsilon_{ijk}\epsilon_{lmn} = \begin{vmatrix} \delta_{il} & \delta_{im} & \delta_{in} \\ \delta_{jl} & \delta_{jm} & \delta_{jn} \\ \delta_{kl} & \delta_{km} & \delta_{kn} \end{vmatrix} = \begin{matrix} \delta_{il}(\delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}) + \\ \delta_{im}(\delta_{jn}\delta_{kl} - \delta_{jl}\delta_{kn}) + \\ \delta_{in}(\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}) \end{matrix} \tag{6.A.2}$$

which can be easily reduced when there is a repeated index (e.g. $i = l$):

$$\epsilon_{ijk}\epsilon_{imn} = \delta_{jm}\delta_{kn} - \delta_{jn}\delta_{km}. \tag{6.A.3}$$

It is worth noting that, in this text, we are using Einstein notation, where a double index indicates the presence of a sum in such index:

$$\epsilon_{ijk}\epsilon_{imn} \equiv \sum_i \epsilon_{ijk}\epsilon_{imn}. \tag{6.A.4}$$

Appendix 6.B Uncertainty principle applied to the angular momentum

We begin this appendix by recalling the Heisenberg uncertainty principle (for position and linear momentum):

$$\Delta x \Delta p \geq \frac{\hbar}{2}. \tag{6.B.1}$$

This result says that we cannot know, simultaneously and with absolute precision, the position of the particle and its linear momentum. Schrödinger generalized this result, this being the Schrödinger (or generalized) uncertainty principle.

Objective The goal of this section is to determine the Schrödinger (or generalized) uncertainty principle. We will use the angular momentum as an example of application.

6.B.1 Schwartz inequality

We can consider a state, represented by the vector $|\Psi\rangle$, as the superposition of two other vectors, as in Eq. (3.12):

$$|\Psi\rangle = |n\rangle + \alpha|m\rangle. \tag{6.B.2}$$

The inner product of these vectors must be a real number:

$$\langle\Psi|\Psi\rangle \geq 0 \tag{6.B.3}$$

and, consequently:

$$\langle n|n\rangle + \alpha\langle n|m\rangle + \alpha^*\langle m|n\rangle + \alpha^*\alpha\langle m|m\rangle \geq 0. \tag{6.B.4}$$

Minimizing the left side with respect to α, we obtain:

$$\frac{d}{d\alpha}\langle\Psi|\Psi\rangle = \langle n|m\rangle + \alpha^*\langle m|m\rangle = 0 \quad \Rightarrow \quad \alpha^* = -\frac{\langle n|m\rangle}{\langle m|m\rangle}, \tag{6.B.5}$$

$$\frac{d}{d\alpha^*}\langle\Psi|\Psi\rangle = \langle m|n\rangle + \alpha\langle m|m\rangle = 0 \quad \Rightarrow \quad \alpha = -\frac{\langle m|n\rangle}{\langle m|m\rangle}. \tag{6.B.6}$$

Taking Eqs. (6.B.5) and (6.B.6) into Eq. (6.B.4) we find:

$$\boxed{\langle n|n\rangle\langle m|m\rangle \geq |\langle n|m\rangle|^2} \tag{6.B.7}$$

The result above is known as Schwartz inequality – which is useful to deduce the Schrödinger (or generalized) uncertainty principle.

6.B.2 Schrödinger uncertainty principle

Let's begin this subsection with the product of the variance of two physical quantities A and B:

$$\begin{aligned}\mathrm{var}(A)\mathrm{var}(B) &= (\Delta A)^2(\Delta B)^2 \\ &= \langle(\hat{A} - \langle\hat{A}\rangle)^2\rangle\langle(\hat{B} - \langle\hat{B}\rangle)^2\rangle,\end{aligned} \tag{6.B.8}$$

which can be rewritten as follows:

$$(\Delta A)^2(\Delta B)^2 = \langle\Psi|(\hat{A}-\langle\hat{A}\rangle)(\hat{A}-\langle\hat{A}\rangle)|\Psi\rangle\langle\Psi|(\hat{B}-\langle\hat{B}\rangle)(\hat{B}-\langle\hat{B}\rangle)|\Psi\rangle, \tag{6.B.9}$$

where we are considering that operators $\hat{A}$ and $\hat{B}$ as Hermitian. We now define:

$$|a\rangle = \delta\hat{a}|\Psi\rangle = (\hat{A}-\langle\hat{A}\rangle)|\Psi\rangle, \tag{6.B.10}$$

$$|b\rangle = \delta\hat{b}|\Psi\rangle = (\hat{B}-\langle\hat{B}\rangle)|\Psi\rangle, \tag{6.B.11}$$

which can be taken into Eq. (6.B.9), yielding:

$$(\Delta A)^2(\Delta B)^2 = \langle a|a\rangle\langle b|b\rangle \geq |\langle a|b\rangle|^2. \tag{6.B.12}$$

Above, we have used Eq. (6.B.7) – Schwartz inequality. Let's now explore the inner product $\langle a|b\rangle$:

$$\langle a|b\rangle = \langle\Psi|\delta\hat{a}\,\delta\hat{b}|\Psi\rangle = \langle\delta\hat{a}\,\delta\hat{b}\rangle, \tag{6.B.13}$$

where $\delta\hat{a}\,\delta\hat{b}$ can be rewritten as follows:

$$\delta\hat{a}\,\delta\hat{b} = \frac{1}{2}[\delta\hat{a},\delta\hat{b}] + \frac{1}{2}\{\delta\hat{a},\delta\hat{b}\} \tag{6.B.14}$$

being $\{\delta\hat{a},\delta\hat{b}\} = \delta\hat{a}\,\delta\hat{b} + \delta\hat{b}\,\delta\hat{a}$ the anti-commutator. We can also write:

$$[\delta\hat{a},\delta\hat{b}] = [(\hat{A}-\langle\hat{A}\rangle),(\hat{B}-\langle\hat{B}\rangle)] = [\hat{A},\hat{B}] \tag{6.B.15}$$

and, since the operators $\hat{A}$ and $\hat{B}$ are Hermitian, we obtain:

$$[\hat{A},\hat{B}] = i\hat{C}. \tag{6.B.16}$$

Check! Eqs. (6.B.15) and (6.B.16) into Eq. (6.B.14) results in:

$$\delta\hat{a}\,\delta\hat{b} = \frac{i}{2}\hat{C} + \frac{1}{2}\{\delta\hat{a},\delta\hat{b}\}. \tag{6.B.17}$$

Eq. (6.B.12), however, requires the squared modulus of the inner product $\langle a|b\rangle$, which can be written as follows:

$$\begin{aligned} |\langle a|b\rangle|^2 &= |\langle\delta\hat{a}\,\delta\hat{b}\rangle|^2 \\ &= \left|\frac{i}{2}\langle\hat{C}\rangle + \frac{1}{2}\langle\{\delta\hat{a},\delta\hat{b}\}\rangle\right|^2 \\ &= \frac{1}{4}|\langle[\hat{A},\hat{B}]\rangle|^2 + |\mathrm{cov}(A,B)|^2 \end{aligned} \tag{6.B.18}$$

where

$$\mathrm{cov}(A,B) = \frac{1}{2}\langle\{\delta\hat{a},\delta\hat{b}\}\rangle \tag{6.B.19}$$

is defined as the covariance. We can further explore Eq. (6.B.19):

$$\begin{aligned}\mathrm{cov}(A,B) &= \frac{1}{2}\langle[(\hat{A}-\langle\hat{A}\rangle)(\hat{B}-\langle\hat{B}\rangle)+(\hat{B}-\langle\hat{B}\rangle)(\hat{A}-\langle\hat{A}\rangle)]\rangle\\ &= \frac{1}{2}\langle\{\hat{A},\hat{B}\}\rangle - \langle\hat{A}\rangle\langle\hat{B}\rangle. \end{aligned}\tag{6.B.20}$$

To continue the development, we take Eq. (6.B.18) into Eq. (6.B.12) and find the Schrödinger (or generalized) uncertainty principle:

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \frac{1}{4}|\langle[\hat{A},\hat{B}]\rangle|^2 + |\mathrm{cov}(A,B)|^2} \tag{6.B.21}$$

This result can be written in a more formal manner. Note that on the left side of the inequality (6.B.21) we have the variances of the quantities A and B; while on the right side, the covariances of these quantities. Thus, organizing terms, we find:

$$\boxed{\det\Xi \geq \frac{1}{4}|\langle[\hat{A},\hat{B}]\rangle|^2} \tag{6.B.22}$$

where

$$\Xi = \begin{pmatrix} \mathrm{var}(A) & \mathrm{cov}(A,B) \\ \mathrm{cov}(A,B) & \mathrm{var}(B) \end{pmatrix} \tag{6.B.23}$$

is the covariance matrix. Eq. (6.B.22) is therefore another way to represent the Schrödinger (or generalized) uncertainty principle.

6.B.3 Application: angular momentum

In this subsection we will use the Schrödinger uncertainty principle for the components of angular momentum. In this case, Eq. (6.B.21) can be rewritten as follows:

$$(\Delta L_u)^2(\Delta L_v)^2 \geq \frac{1}{4}|\langle[\hat{L}_u,\hat{L}_v]\rangle|^2 + |\mathrm{cov}(L_u,L_v)|^2. \tag{6.B.24}$$

Based on this chapter, we know that:

$$\begin{aligned}&\langle\hat{L}_z\rangle = m\hbar, \qquad \langle\hat{L}_x\rangle = \langle\hat{L}_y\rangle = 0, \qquad \langle\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x\rangle = 0, \quad \text{and}\\ &\langle\hat{L}_z\hat{L}_x\rangle = \langle\hat{L}_y\hat{L}_z\rangle = 0.\end{aligned}\tag{6.B.25}$$

Thus, for any component of the angular momentum:

$$\mathrm{cov}(L_u, L_v) = \frac{1}{2}\langle\{\hat{L}_u, \hat{L}_v\}\rangle - \langle\hat{L}_u\rangle\langle\hat{L}_v\rangle = 0. \tag{6.B.26}$$

In this case, the uncertainty relation for the components of angular momentum can be written as (using Eq. (6.4)):

$$\Delta L_u \, \Delta L_v \geq \frac{\hbar}{2}|\epsilon_{uvw}\langle\hat{L}_w\rangle|. \tag{6.B.27}$$

Based on the above result, we have $\Delta L_z \, \Delta L_x = \Delta L_y \, \Delta L_z \geq 0$. This means that the standard deviation of the z component of the angular momentum can be zero (and indeed it is – check it out!), which does not violate the uncertainty principle. We can also conclude that:

$$\Delta L_x \, \Delta L_y \geq \frac{\hbar^2}{2}|m|. \tag{6.B.28}$$

To think What is the physical meaning of the above result?

Appendix 6.C Exercises

Conceptual exercises

Exercise 6.1. What is the physical meaning of the commutation relation $[\hat{L}_u, \hat{L}_v] = i\hbar\epsilon_{uvw}\hat{L}_w$?

Exercise 6.2. Discuss the variance of the components of angular momentum.

Text-based exercises

Exercise 6.3. Verify that the components of a cross product can be written in terms of the Levi-Civita symbol.

Exercise 6.4. Verify the following relationships:

1. $[\hat{L}^2, \hat{L}_x] = [\hat{L}^2, \hat{L}_y] = 0$ (Theorem 6.1).
2. $[\hat{L}_-, \hat{L}_+] = -2\hbar\hat{L}_z$ (Theorem 6.2).
3. $[\hat{L}_z, \hat{L}_-] = -\hbar\hat{L}_-$ (Theorem 6.3).
4. $\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$.
5. $a = b_{min}(b_{min} - \hbar)$ (Corollary 6.5).

Practice exercises

Exercise 6.5. Show that $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$, considering that $\hat{L}_x = (\hat{y}\hat{p}_z - \hat{z}\hat{p}_y)$ and $\hat{L}_y = (\hat{z}\hat{p}_x - \hat{x}\hat{p}_z)$.

Exercise 6.6. Verify the angular momentum algebra (Eqs. (6.4), (6.16), (6.20), (6.22), (6.24), (6.48), (6.49), and (6.58)), through the matrix form of the angular momentum operator, for $l = 1$.

Exercise 6.7. Determine the matrices that represent the operators $\hat{L}_x$, $\hat{L}_y$, $\hat{L}_z$, and $\hat{L}^2$, for $l = 3/2$.

Exercise 6.8. Based on the previous exercise, show that the components of angular momentum are Hermitian operators, while the raising and lowering operators are non-Hermitian.

Exercise 6.9. Consider an angular momentum $l = 1/2$ and its component $\hat{L}_x$.

1. What are the eigenvalues?
2. What are the corresponding vectors?
3. What is the relationship between the vectors $|\pm\rangle_x$ and $|\pm\rangle$?
4. What is the matrix for the basis change?
5. Use the basis change matrix to diagonalize $\hat{L}_x$.

Exercise 6.10. Repeat the previous exercise, considering $\hat{L}_y$.

Development exercises

Exercise 6.11. Consider an angular momentum $l = 1$ and the operator $\hat{G} = \hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x$.

1. What are the eigenvalues?
2. What are the corresponding vectors?
3. What is the relationship between the vectors of $\hat{G}$ and $\hat{L}_z$?
4. What is the matrix for the basis change?
5. Use the change of basis matrix to diagonalize $\hat{G}$.

Exercise 6.12. Determine the cross product $\vec{\hat{L}} \times \vec{\hat{L}}$.

Exercise 6.13. Consider the operator $\hat{L}^{-2}$.

1. Obtain $[\hat{L}^2, \hat{L}^{-2}]$.
2. Obtain the eigenvalues and vectors of this operator.

Exercise 6.14. Consider the commutator $[\hat{r}^2, \hat{L}_u]$, where u is some direction in space.

1. Using the Levi-Civita symbol, obtain the result of this commutator.

2. Is the result obtained in the previous item expected?

Exercise 6.15. Use the Levi-Civita symbol to determine the following commutators:

1. $[\hat{L}_u, \hat{p}_v]$.
2. $[\hat{L}_u, \hat{x}_v]$.
3. $[\vec{\hat{A}} \cdot \vec{\hat{L}}, \vec{\hat{B}} \cdot \vec{\hat{L}}]$, where the components of the vectors $\vec{\hat{A}}$, $\vec{\hat{B}}$ and $\vec{\hat{L}}$ commute with each other.

Exercise 6.16. Make a vector diagram of the angular momentum, with its magnitude and projection along the z-axis. From this figure, make an estimate of the projection of this vector on the xy-plane and compare it with the standard deviation of the quantum case.

Exercise 6.17. Consider a particle with angular momentum described by the vector:

$$|\Psi\rangle = \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}. \tag{6.C.1}$$

Calculate the probability $P(l,m)$ of obtaining each of the possible angular momentum eigenvalues after a measurement of $\hat{L}_x$.

Exercise 6.18. Using the algebra of angular momentum, write analytically the spherical harmonic corresponding to $(l,m) = (2,+2)$.

Exercise 6.19. Calculate the expectation values: $\langle \hat{L}_x \hat{L}_y \rangle$ and $\langle \hat{L}_y \hat{L}_x \rangle$.

Exercise 6.20. Calculate the expectation values: $\langle \hat{L}_x \hat{L}_z \rangle$ and $\langle \hat{L}_y \hat{L}_z \rangle$.

Advanced exercises

Exercise 6.21. An experiment with an electron performs a measurement of the spin projection along the z-axis and finds $m = +1/2$ (in units of $\hbar$). Subsequently, another experiment is conducted, which measures the spin projection of this particle along an arbitrary direction, given by:

$$\hat{n} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z} \tag{6.C.2}$$

1. Write the matrices $\hat{S}_x$, $\hat{S}_y$ and $\hat{S}_z$ in the $\{|l,m\rangle\}$ basis.
2. Write the spin operator $\hat{S}_n$.
3. Obtain the spin eigenvalues after the measurement in the direction $\hat{n}$.
4. Obtain the corresponding vectors.
5. What is the probability $P(m)$ of obtaining each of the eigenvalues after a spin measurement in the direction $\hat{n}$?

Use the trigonometric relations: (i) $\sin x = 2\sin(x/2)\cos(x/2)$, (ii) $1+\cos(2x) = 2\cos^2 x$, and (iii) $1-\cos(2x) = 2\sin^2 x$.

Exercise 6.22. Consider that the angular part of a certain wave function is given by:

$$A(x,y,z) = \frac{\alpha x}{\sqrt{x^2+y^2+z^2}}, \tag{6.C.3}$$

where α is a constant. Obtain the probability $P(l,m)$ of finding the particle with angular momentum $(l,m) = (1,\pm 1)$ and $(1,0)$.

Exercise 6.23. Consider a system described by the Hamiltonian of a three-dimensional harmonic oscillator:

$$\hat{\mathcal{H}} = \frac{\hat{p}^2}{2m} + \frac{m}{2}\omega^2 r^2 \tag{6.C.4}$$

1. Write $\hat{\mathcal{H}}$ in terms of the angular momentum operator and radial contributions only.
2. From the previous result, calculate $[\hat{\mathcal{H}}, \hat{L}_z]$.